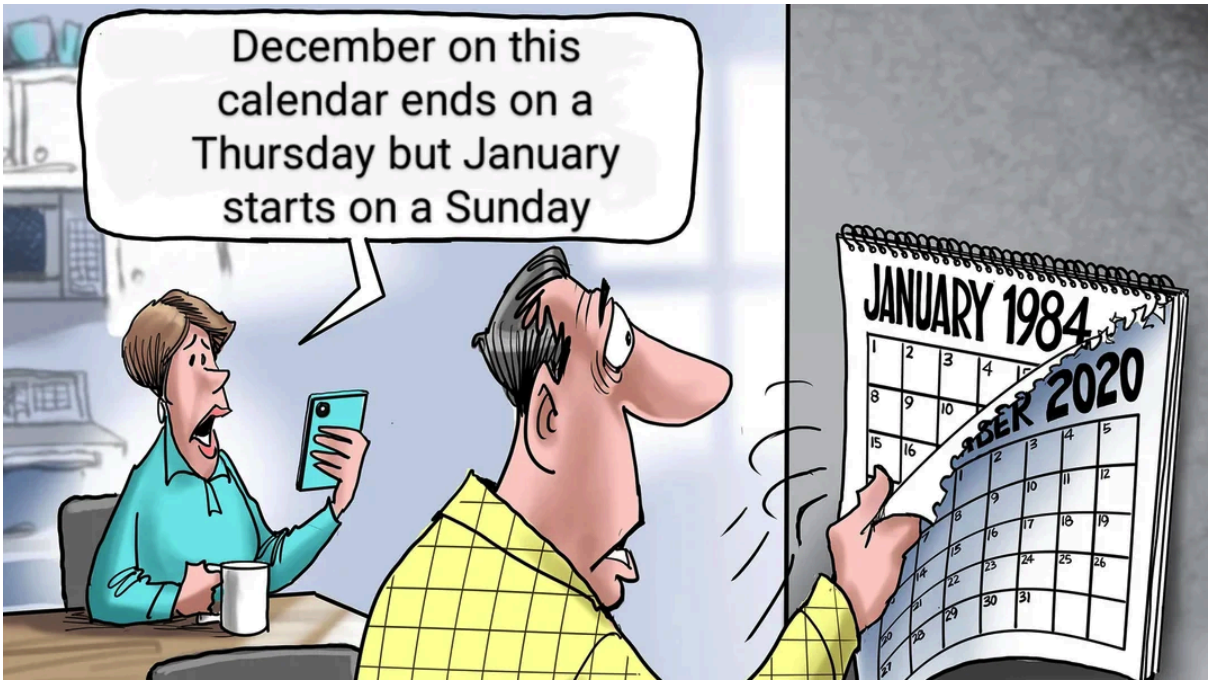
The Calendar Strategy

Profiting From The Relative IV Across Expirations.



Prepared by Sean Ryan

Exclusively For Predicting Alpha Annual Members

Because you guys rock 😎

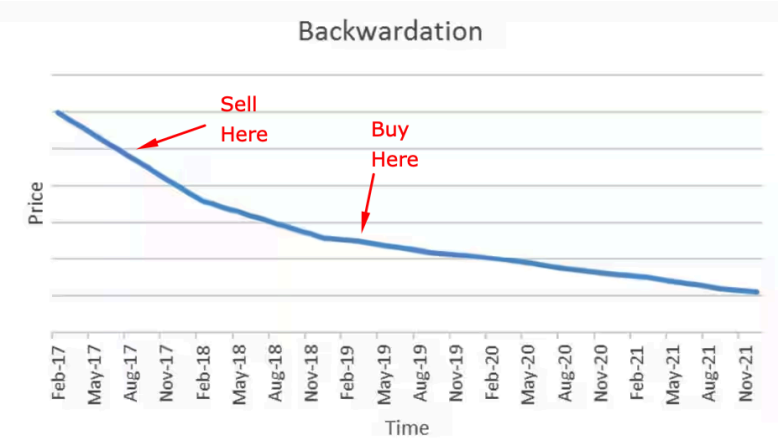
Table of Contents

Part 1: Strategy Introduction + Overview **3**
What is the Calendar Spread Strategy? 3
Why Does This Strategy Make Money? 3
Strategy Performance Snapshot 3
What's The Signal? 4
What's The Trade? 4
Which Tickers Should You Start With? 5
Is This Strategy Different from Other Premium Collection Strategies? 5
What are the primary challenges with this strategy? 5
Is There Any Benefit to Diversifying Across Trades? 6
Part 2: How Calendar Spreads Work **6**
Trade Definition 7
Greeks & Market Exposures 7
A Bet on Forward Volatility 8
Relative Value Trade 9
Impact of Term Structure Changes 11
Understanding Term Structure Movements 12
Exploiting Term Structure Deviations 14
Part 3: Research & Strategy Performance **15**
Download The Strategy Research Paper 15
Summary of the Strategy Research 15
Forward Factor as the Signal 16
Decile Analysis Results 16
Cutoff for Decile 9 17
Impact of Bid-Ask Spread 17
Impact of Earnings 18
Impact of Company Size 19
Optimal Trade Summary 19
Part 4: How to Run The Strategy **21**
Step 1: Create a List of Potential Opportunities 21
Step 2: Analyze the Data 22
Step 3: Check What is Driving This Forward Factor 23
Step 4: Pick Your Live Strikes and Expirations 25
Step 5: Price The Forward Factor 26
Step 6: Get a Fill On The Trade! 28
Step 7: Managing The Trade 28
Conclusion **29**

Part 1: Strategy Introduction + Overview

What is the Calendar Spread Strategy?

Buy calendar spreads when the Term Structure for stocks is in backwardation that is steeper than it should be.



Why Does This Strategy Make Money?

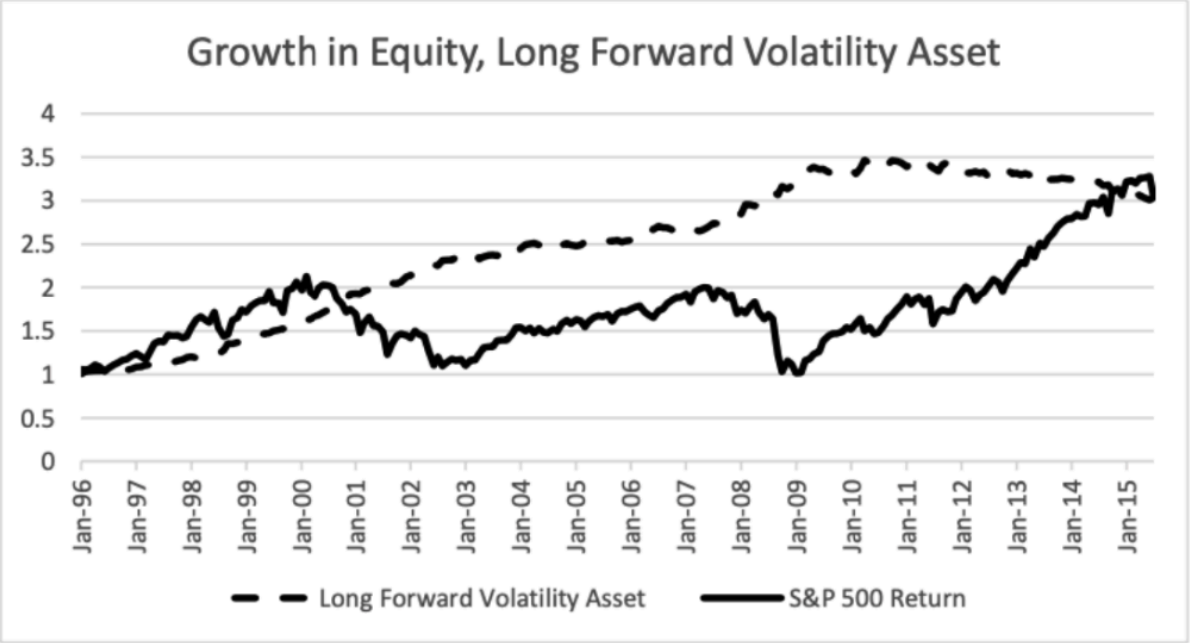
This strategy makes money primarily because we are selling expensive short dated option contracts. It is the “short leg” of our calendar spread that is really driving our returns. The reason that we structure the trade as a calendar spread is because the Forward Factor (the signal we use to find trades) helps us identify opportunities where we can buy a relatively cheap longer dated option as a hedge. This allows us to put on option selling trades that are hedged without giving up as much of our expected value.

As you will see in the research, this strategy allows us to carry short volatility positions with very cost effective hedges, results in a much smoother PnL and allows us to avoid many of the drawdowns that typically occur during high volatility environments (ie: market crashes, etc). Essentially, we are being paid to **sell expensive volatility** while maintaining protection through the longer expiration option.

The causes for these opportunities can vary, but from actively trading this I’ve seen that these opportunities usually follow one-off larger price movements or uncertainties. Think buyout rumors, product releases, regulatory changes, etc.

Strategy Performance Snapshot

Here is a backtest of what running this strategy on the best performing basket of tickers (covered in the research section of this course) compared against SPX returns.



As you can see, the returns for this strategy are very smooth. Since long calendar spreads are debit positions, they are more insulated from market panics and large moves, which results in a very smooth PnL when we can put together a basket of trades that each have positive expectancy.

What's The Signal?

The research shows that opportunities with a **Forward Factor greater than 16% (0.16)** were the best for trading long calendar spreads, because it indicates that front month volatility is most expensive relative to back month volatility

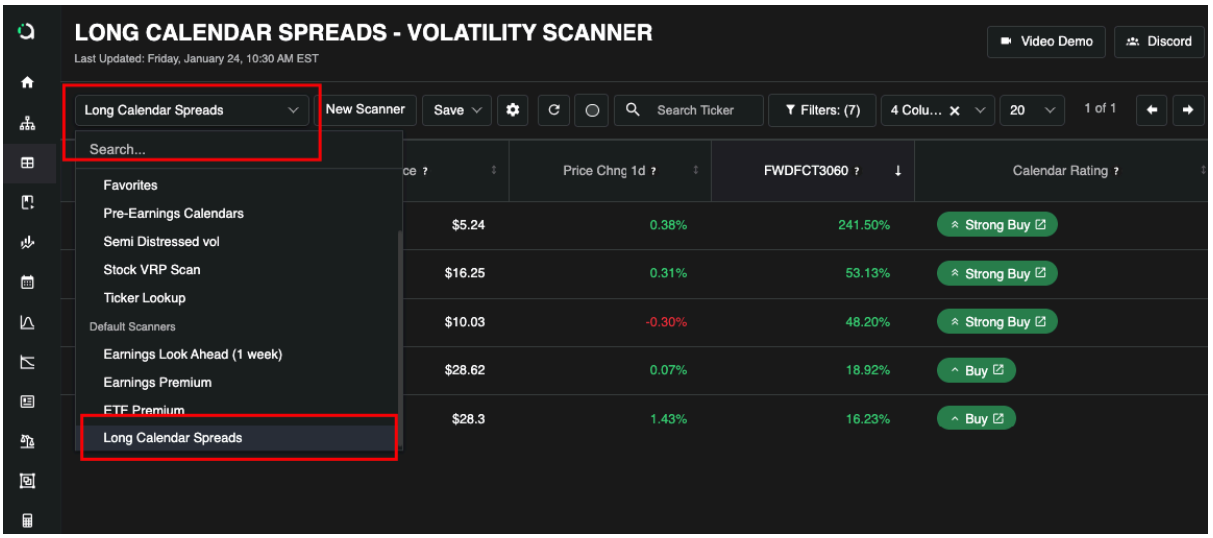
---

What's The Trade?

When we find an opportunity, we sell close expiration options (7-30 DTE) and buy longer expiration options (20-60 DTE). This creates a structure called a long calendar spread.

Which Tickers Should You Start With?

In the Predicting Alpha Terminal there is a scanner called “Long Calendar Spreads” pre built for you. You can access it using the dropdown in the top left corner of the scanner. The tickers on this list are good opportunities for the 30DTE/60DTE calendars and represent opportunities that are good to start with.



This is where you access the Long Calendar Spreads scan that is pre built for you in the Terminal

Is This Strategy Different from Other Premium Collection Strategies?

Yes it’s completely different! The premise for this strategy is that option prices across time can be cheap/expensive relative to each other *within the same ticker*. It’s actually a relative value trade, where we aim to capture the risk premium in short dated options while hedging with cheaper, longer dated options. It’s a great strategy to run alongside the core premium selling strategies we focus on.

What are the primary challenges with this strategy?

The reason this strategy is considered a **bonus** rather than one of our core strategies (like ETF or Earnings) is because it requires a more hands-on approach. You will need to:

- Conduct deeper research to understand the context around the tickers you analyze.
- Have a strong understanding of trade execution since calendar spreads tend to be less liquid, meaning **getting good fills** is critical for profitability.

Is There Any Benefit to Diversifying Across Trades?

Yes there is a significant benefit to diversifying across many trades. At the end of the day, just like with our other strategies, the edge in each position is quite small. The way to achieve the most forecastable returns is to take many trades.

Now that we have covered the basics of this strategy, let's talk about how calendar spreads actually work so you can feel confident executing these trades.

Part 2: How Calendar Spreads Work

Even if you have traded a calendar spread before, make sure to go through this section. Calendars are more complex than most of us have been taught over the years, and in this module we are going to go over exactly how they work and why they are the perfect structure to trade for this strategy.

Trade Definition

A calendar spread is an options trading strategy where you buy and sell the same strike option across two different expiration dates.

A long calendar spread (which we use in this strategy) involves selling the option with the closer expiration date and buying the option with the later expiration date. For both expirations, you trade the same strike. For instance, a long calendar spread example would be selling an $XYZ company July 150 strike call and buying a September 150 strike call. Here is an example of what a basic long calendar spread looks like:

When you trade a long calendar spread, it is a debit position so the maximum amount you can lose is the amount that you paid. In the example above, the maximum loss is $3.40, or $340/lot.

Greeks & Market Exposures

It is crucial to start by understanding how a calendar spread's profit and loss (PnL) will react in various market scenarios. This involves analyzing the Greeks and exposures associated with the structure.

Initial Exposures of an ATM Calendar Spread:

Delta Neutral: At the inception of the trade, we are indifferent to the stock's price direction.
Short Gamma: Rapid price movements can negatively impact our position.
Long Theta: All else being equal, the passage of time will generate profit for us.
Long Vega*: An increase in implied volatility will benefit our position.
*Traders will often assume that this position is long vega. This is a common mistake that we will break down in a little bit.

At first glance, a calendar spread appears to be profitable if the market remains stable or if there is a significant increase in implied volatility. It appears likely to incur losses if the stock experiences a sharp movement in its share price.

But it's important to remember that the key component of a calendar spread which makes it different from other trades is that it involves trading *across multiple expirations*, which adds a level of complexity (or opportunity) to how it can change in value.

A Bet on Forward Volatility

This is where many traders get confused.

A calendar spread is essentially a relative value trade between gamma and vega.

Think of it as "*I'm short gamma and hedging with vega*."

Alternatively, it can be viewed as a bet on "forward volatility." (The core concept of this strategy).

But what exactly is forward volatility?

Consider this scenario:

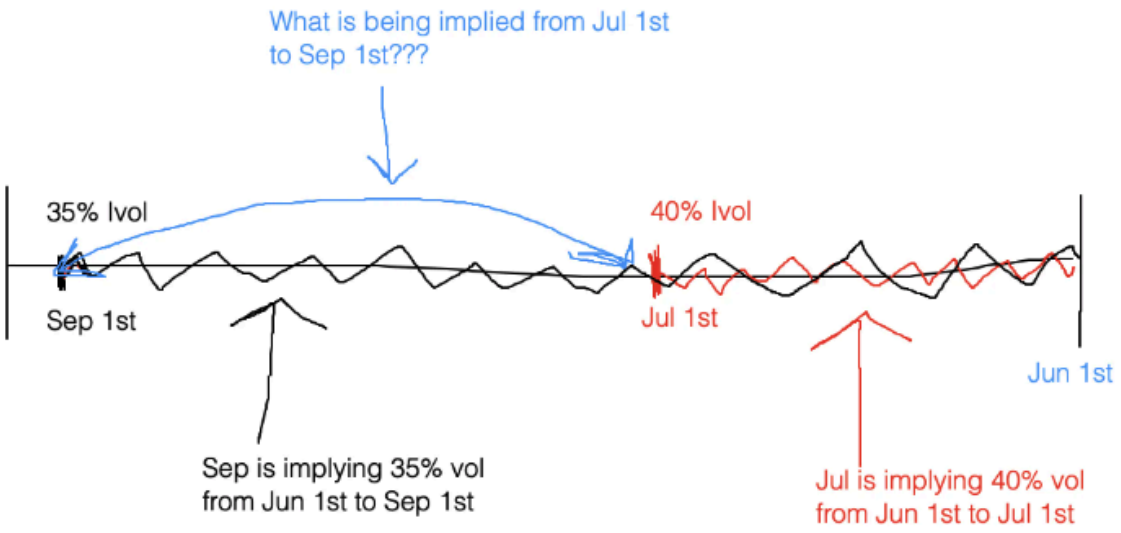
Today is June 1st, and the implied volatilities are as follows:

- July 1st Expiration: 40%
- September 1st Expiration: 35%

What conclusions can we draw from these implied volatilities?

We understand that over the next 30 days, until the July 1st expiration, the market is implying a 40% volatility. Similarly, over the next 90 days, until the September 1st expiration, the market is implying a 35% volatility.

One question that we do not have the answer to is what the volatility between July 1st and September 1st is expected to be. The volatility *between* the expirations.



The volatility between expirations is what we refer to as the *forward volatility*. This is calculated for you in the Predicting Alpha Terminal and is a critical component of creating the signal for this strategy. In this example, the 30/90 forward volatility would be 32%.

When trading a calendar spread, you are essentially expressing a view on forward volatility. If you bought this calendar spread, you would be purchasing forward volatility at 32%.

Relative Value Trade

It's **crucial** to start thinking of a calendar spread as a relative value play between gamma and vega.

This next concept might be tricky, but it's essential to fully understand to run this strategy effectively.

When trading the July/September Calendar, the July expiration will have significantly more gamma than the September expiration, meaning it is more sensitive to rapid changes in the underlying share price. Conversely, the September expiration will have more vega, meaning it is more sensitive to changes in the level of implied volatility.

We sold July options at a 40% implied volatility and bought September options at a 35% implied volatility, thereby locking in a 32% forward volatility.

Now, imagine that over the next 30 days, the realized volatility is 40%.

This scenario would imply a break-even point for our July expiration.

The Tricky Question: What Will September Be Trading At?

Here's the challenging question: what will the September implied volatility be?

If the September expiration is trading higher than 32%, we will have profited from the calendar spread. If it is trading below 32%, we will have incurred a loss. That is because when the July option expires, and all that is left is the time from the July expiration to the September expiration, the implied volatility for the September expiration should theoretically be equal to the forward volatility we locked in. If it's higher, since we are long vega in the back month, we will have made money on the September contracts. If lower, we will have lost money.

What If Realized Volatility Is High In The Short Term?

If the realized volatility over the next 30 days is 60%, we will have lost on the July expiration since we sold at a 40% volatility level. However, the September expiration should be trading at a much higher volatility, correct?

A significant portion of the PnL for July comes from gamma, whereas most of the PnL for September comes from vega.

What If Realized Volatility Is Low In The Short Term?

Let's consider another scenario:

Imagine the realized volatility over the next 30 days is 10%.

In this case, we will profit significantly on our July expiration since we sold at a 40% volatility and realized only 10% volatility.

But what about the September expiration?

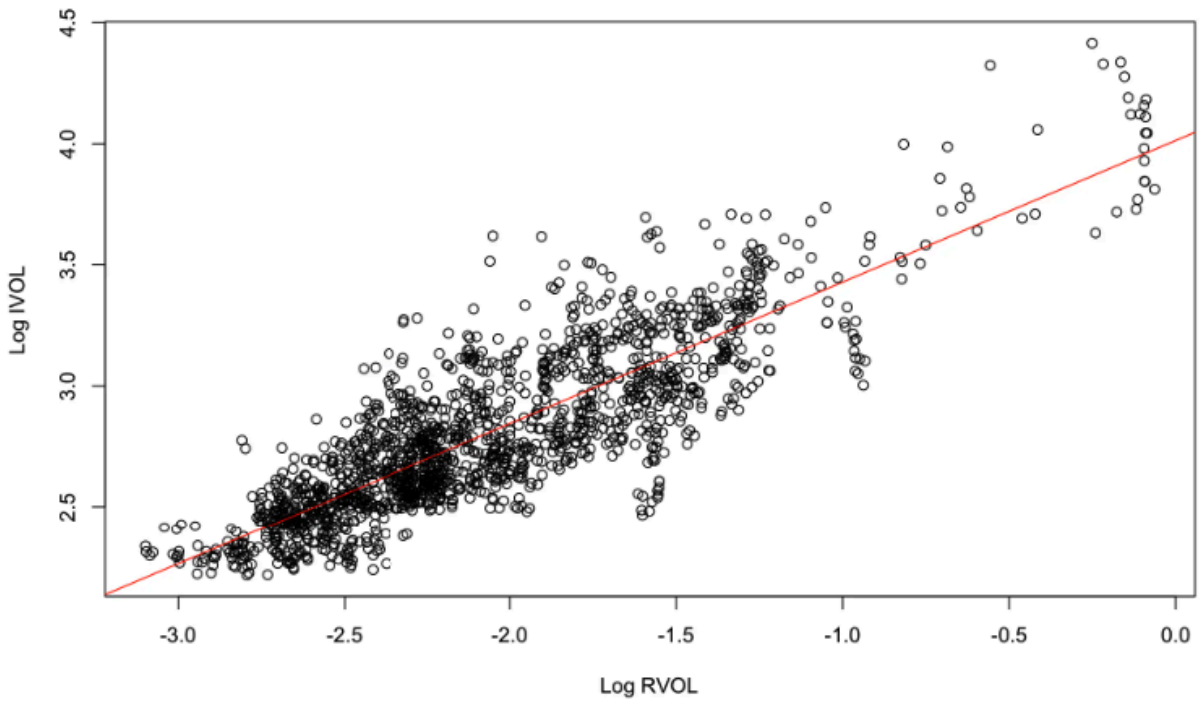
If we are only realizing 10% volatility, the implied volatility for September will decrease substantially.

Therefore, in this example, we made money on our gamma leg and lost money on our vega leg.

The graph below illustrates the relationship between implied volatility (IVOL) and realized volatility (RVOL) for the QQQ.

This strong correlation indicates that it is uncommon to lose money on both the gamma and vega legs or to profit from both.

When RVOL is high, IVOL tends to be high as well, and vice versa. Instead, you are trading the relative value or “richness” of one leg compared to the other.

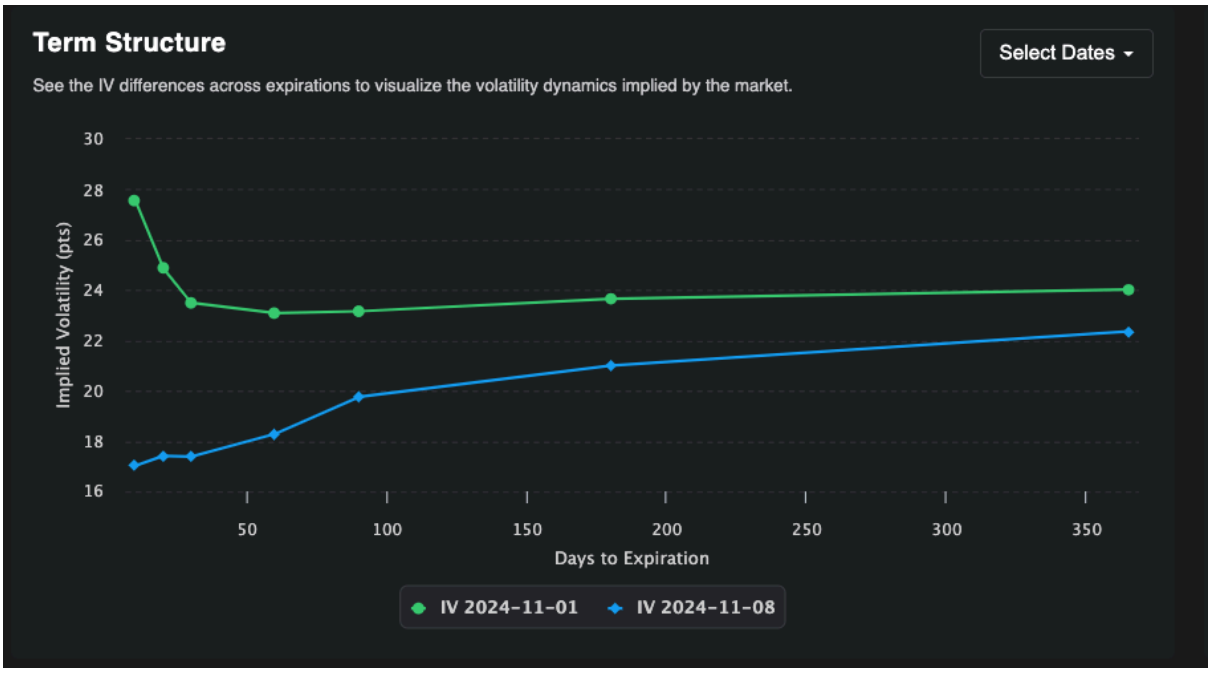


This is why calendars can be attractive plays. They are hedged to some degree because you have a position on two things that typically have inverse PnLs from each other. Since their PnLs operate in this way, they are typically priced fairly against each other. However, there can be opportunities that come from changes in the term structure where the relative cheapness/expensiveness of the front and back month options can change, creating an ideal opportunity to trade a calendar spread (the core of this strategy.

Impact of Term Structure Changes

To identify viable trade ideas for calendar spreads, we need to understand how the term structure moves.

The term structure describes the pattern of implied volatility across different expirations or tenors. Below is an example of AAPL's term structure on two different days. You can observe that the 30-day options are trading at 25%, and the 90-day options are trading at 23%. Additionally, over a one-week period, the volatility across the term structure has decreased to roughly 17% and 20%.



You can see how as the expected volatility for a ticker changes, the whole term structure reacts. When volatility is expected to be high, we can see this change reflect in all of the expirations to some degree. When it comes back down, that change should also reflect across all the expirations. *The degree of change across expirations* is what can lead to opportunity, and it's what we will discuss now.

Understanding Term Structure Movements

It's crucial to grasp how the term structure "usually" moves, which is typically in a "root time" fashion. By "root," we mean the square root.

In essence, this means that short-dated options are more sensitive than longer-dated options. Sensitivity here refers to how much they move in terms of implied volatility.

The Best Way to Explain: An Example

Let's illustrate this with an example of a flat term structure:

30-day IVOL = 30%
60-day IVOL = 30%
90-day IVOL = 30%
120-day IVOL = 30%
365-day IVOL = 30%

In this scenario, all the implied volatilities across different expirations are equal at 30%.

Explaining with an Example: Volatility Shock

Let's consider a scenario where a significant event occurs, such as China deciding to halt international trade with the USA. This event would likely cause an increase in volatility.

If the 365-day implied volatility (IVOL) jumps by 10 points to 40% implied volatility, we need to determine how much the other expirations should change.

The answer lies in the concept of "root time." Here's how it works:

A 1-year volatility change of 10 points translates to changes in shorter-term volatilities by the square root of the ratio of days in a year to days until expiration, multiplied by the annualized volatility change.

$$\text{New IVOL} = \sqrt{\left(\frac{\text{Days in a year}}{\text{Days to expiration}}\right)} \times \text{Annualized vol change} + \text{Original IVOL}$$

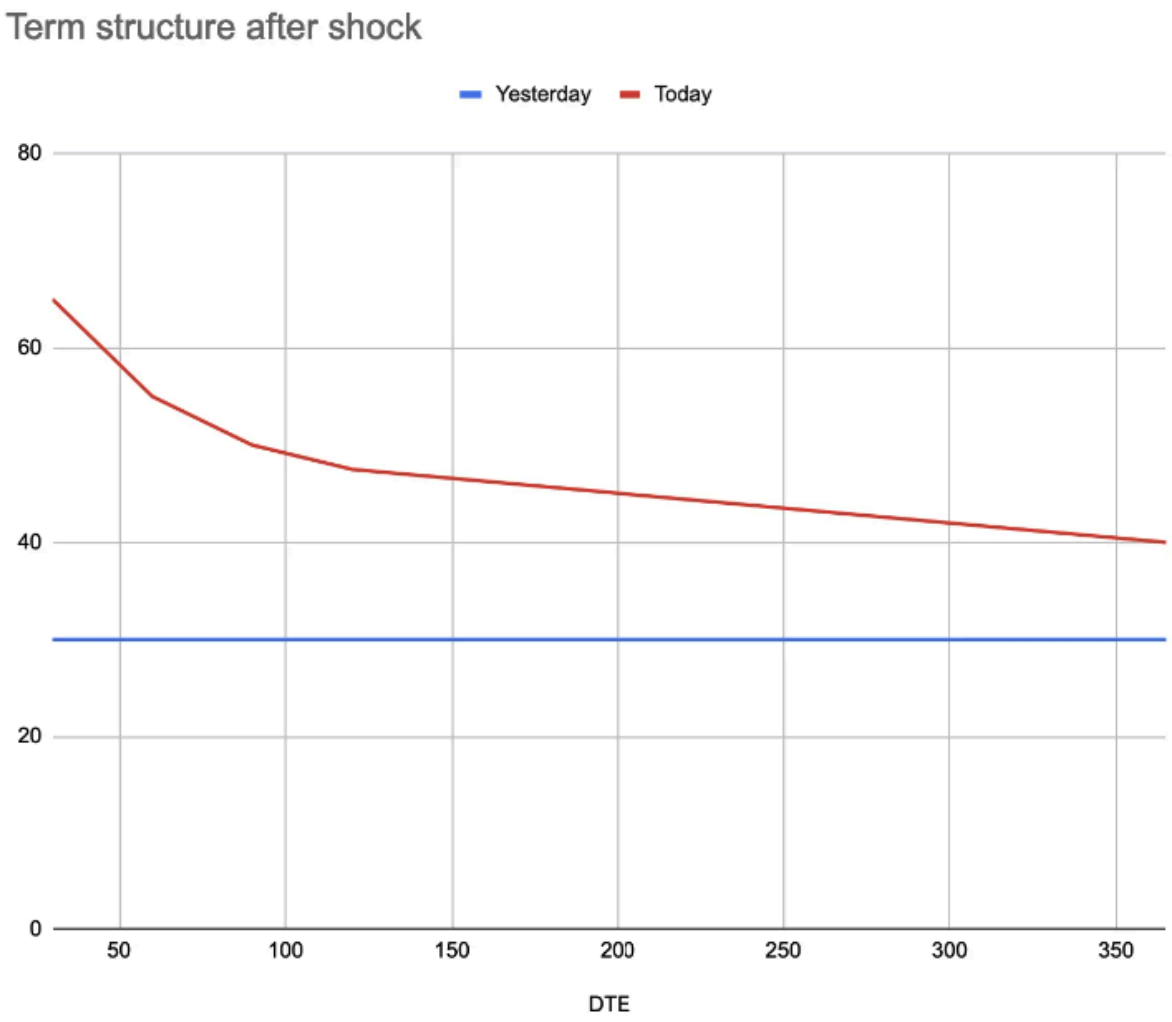
Here are the calculated multipliers:

- $\sqrt{\left(\frac{365}{30}\right)} = 3.48$
- $\sqrt{\left(\frac{365}{60}\right)} = 2.46$
- $\sqrt{\left(\frac{365}{90}\right)} = 2.01$
- $\sqrt{\left(\frac{365}{120}\right)} = 1.74$
- $\sqrt{\left(\frac{365}{365}\right)} = 1$

Using these multipliers, the new volatilities are:

- **30-day IVOL:** $3.48 \times 10 + 30 = 65\%$ (a 35-point increase)
- **60-day IVOL:** $2.46 \times 10 + 30 = 55\%$ (a 25-point increase)
- **90-day IVOL:** $2.01 \times 10 + 30 = 50\%$ (a 20-point increase)
- **120-day IVOL:** $1.74 \times 10 + 30 = 47.5\%$ (a 17.5-point increase)
- **365-day IVOL:** $1 \times 10 + 30 = 40\%$ (a 10-point increase)

This is what the change in the term structure would look like:



Exploiting Term Structure Deviations

Traders can exploit deviations from root time movements. For instance, if the 90-day IVOL only moved to 45% after the shock, when it should have theoretically moved to 50%, they would buy options with that expiration and sell options with surrounding expirations using a calendar spread.

Simply put, when we can find times where the volatility for different expirations has moved more or less relative to the other expirations, we can structure a relative value trade using a calendar spread to express a view on this! The forward factor from Jim Campasano measures the relative difference in implied volatility between a short-term and long-term expiration. When the front expiration's implied volatility is disproportionately high relative to the back expiration, it signals a deviation from the expected term structure behavior.

As you will see in the research, when we can put together a portfolio of trades with these term structure characteristics, we are able to generate a very attractive and smooth return for ourselves.

Part 3: Research & Strategy Performance

Download The Strategy Research Paper

The idea for this strategy comes from research by Jim Campasano. We originally read this back in 2022 and members have been trading it since then. This lesson provides you with the research that went into creating the signal for this strategy, so that you can review it for yourself.

To examine the original research for yourself, download it here:
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3240028

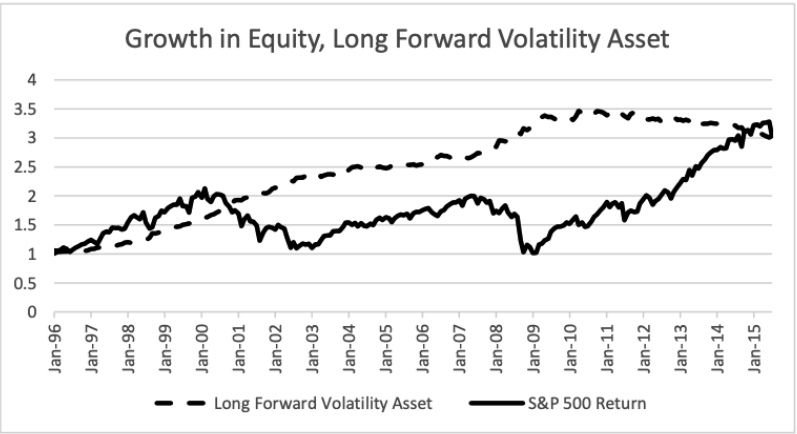
Summary of the Strategy Research

The research examined the predictability of equity implied volatility using the term structure. In the research Jim Campasano uses something called a **Forward Volatility Asset (FVA)** as the trade he placed to prove the effect exists, which is essentially a 30dte/60dte long calendar spread, where a 2-month ATM option is bought, and a 1-month ATM option is sold. This structure represents the exact strategy being discussed. Portfolios were formed monthly by sorting tickers into deciles based on differences between implied volatility and forward volatility levels, referred to here as the **forward factor (the signal we use to find trades)**

This is what the strategy returns look like

Figure 2: Returns to Long Forward Volatility Asset Portfolio After Transactions Costs

This figure charts the value of a portfolio investing in the S&P 500 Index and the Quintile 5 slope, Quintile 1 bid-ask spread forward volatility asset portfolio after accounting for transactions costs: returns are calculated after applying a 50% effective bid-ask spread. Each portfolio begins with $1. Due to the large monthly returns as compared to the S&P 500 Index, 1% of the portfolio is invested in the FVA portfolio.



Forward Factor as the Signal

The **forward factor** measures the difference between front month implied volatility and forward volatility levels. It serves as the signal to identify opportunities in the strategy. When the forward factor is high (i.e., forward volatility is significantly lower than spot volatility), it creates the conditions for profitable long calendar spreads. Sorting tickers by their forward factor reveals a clear relationship: the higher the forward factor, the higher the returns of a long calendar spread. This is demonstrated through a decile analysis.

Decile Analysis Results

The research specifically evaluates the performance of the 30/60 calendar spread (FVA). Decile 9 and 10 portfolios exhibited significantly higher returns compared to other deciles, benefiting from the most inverted term structures.

Table 5 highlights the returns for each decile, showing the relationship between forward factors and calendar spread profitability.

Table 5: Returns of Forward Volatility Asset Portfolios

Each month, portfolios of forward volatility assets (FVA) are formed based on the implied volatility differential between the two legs, as defined by $\frac{1mIV}{FV(1,1)} - 1$. The FVA consists of a long two month ATM equity option straddle and a short one month ATM equity option straddle. Portfolio 1 (10) holds FVAs whose implied volatility curves have steepest upward (downward) slope. The FVAs are constructed such that the notional amounts of the one month and two month options are equal. 1m refers to the one month straddle returns; 2m refers to the two month straddle returns; and FVA refers to the FVA returns. The period examined spans from January, 1996, to July, 2015, and includes 284,984 FVAs and 6,799 equities.

	Decile 1	2	3	4	5	6	7	8	9	Decile 10	'10-1
1m	-0.0369***	-0.0550***	-0.0594***	-0.0617***	-0.0679***	-0.0758***	-0.0797***	-0.0925***	-0.1011***	-0.1390***	
2m	-0.0220***	-0.0178**	-0.0147	-0.0113	-0.0113	-0.0122	-0.0082	-0.0134	-0.0093	-0.0197***	
FVA	0.0425***	0.0919***	0.1122***	0.1374***	0.1588***	0.1790***	0.2143***	0.2475***	0.3135***	0.5277***	0.4852***
St. Error	(0.0087)	(0.0096)	(0.0101)	(0.0113)	(0.0121)	(0.0117)	(0.0125)	(0.0127)	(0.0138)	(0.0169)	(0.0154)
Sharpe	1.11	2.17	2.52	2.75	2.96	3.46	3.86	4.41	5.13	7.05	7.12

These figures illustrate the signal's strength but do not account for transaction costs (covered in a bit).

Also note that the description above the table explains that the deciles are based on the Forward Factor, which is the equation given in the first sentence.

Cutoff for Decile 9

A **16% forward factor** was the approximate cutoff for inclusion in Decile 9. Note that the forward factor cut off for each decile is given in the first row of the table below.

Table 4: Summary Statistics: Forward Volatility Assets

Each month, portfolios of forward volatility assets (FVA) are formed based on the implied volatility differential between the two legs, as defined by $\frac{1mIV}{FV(1,1)} - 1$. The forward volatilty asset consists of a long two month ATM equity option straddle and a short one month ATM equity option straddle. Portfolio 1 (10) holds FVAs whose implied volatility curves have steepest upward (downward) slope. The period examined spans from January, 1996, to July, 2015, and includes 284,984 FVAs and 6,799 equities.

	Decile 1	2	3	4	5	6	7	8	9	10	All
$\frac{1mIV}{FV(1,1)} - 1$	-0.2038	-0.1083	-0.0665	-0.0346	-0.0058	0.0230	0.0558	0.0974	0.1630	0.4157	0.0417
1m IV	0.3928	0.4038	0.4168	0.4259	0.4364	0.4441	0.4570	0.4681	0.4863	0.5289	0.4425
2m IV	0.4488	0.4280	0.4307	0.4325	0.4366	0.4382	0.4441	0.4469	0.4525	0.4594	0.4374
FV(1,1)	0.4942	0.4494	0.4433	0.4385	0.4368	0.4325	0.4318	0.4261	0.4185	0.3853	0.4305
Delta	0.0050	0.0030	0.0046	0.0056	0.0067	0.0064	0.0070	0.0076	0.0071	0.0064	0.0035
N	28387	28519	28527	28516	28482	28560	28527	28516	28530	28420	284984

Portfolios with forward factors at or above this level experienced favorable risk-adjusted returns and justify focusing on the top two deciles for trading.

Impact of Bid-Ask Spread

Transaction costs, modeled as 50% of the bid-ask spread, significantly reduced returns across most deciles. While **Deciles 9 and 10** remained robustly profitable after transaction costs, the importance of liquidity becomes clear: **the best trades are those in Deciles 9 and 10 that also fall into the top 20% most liquid securities (Spread Quintile 1)**. Securities with narrow bid-ask spreads mitigate the erosion of returns caused by transaction costs, ensuring that the strategy captures the full benefit of high forward factors.

Note: Jim tracked what would have happened if he bought the 30/60 FVA every 30 days at the mid price, but didn't actually have to deal with slippage, liquidity, and other realities of trading. Through the experience of running this strategy live, we've seen that these factors are what make calendar spreads challenging and is why getting good fills is an extremely important part of the strategy.

Table 6: Forward Volatility Asset Portfolio Returns After Transactions Costs

Each month, portfolios of forward volatility assets (FVA) are formed based on the implied volatility differential between the two legs, as defined by $\frac{1mIV}{FV(1,1)} - 1$. The FVA consists of a long two month ATM equity option straddle and a short one month ATM equity option straddle. The FVAs are constructed such that the notional amounts of the one month and two month options are equal. Portfolio 1 (5) holds FVAs whose implied volatility curves have steepest upward (downward) slope. The portfolios are sorted again according to the bid-ask spread of the one month options. The returns shown are calculated using an effective bid-ask spread of 50% that which is quoted. The period examined spans from January, 1996, to July, 2015, and includes 284,984 FVAs and 6,799 equities.

	Term Structure					
Spread	1 (depressed)	2	3	4	5 (inverted)	5-1
1	-0.0663	-0.0363	-0.0036	0.0293***	0.0965***	-0.0705
	(0.0131)	(0.0124)	(0.0136)	(0.0134)	(0.0157)	(0.0197)
Sharpe Ratio	-1.15	-0.66	-0.06	0.49	1.39	-0.80
2	-0.1239	-0.0942	-0.0757	-0.0455	0.0228	-0.2493
	(0.010)	(0.0103)	(0.0114)	(0.0128)	(0.0161)	(0.0164)
3	-0.1708	-0.1490	-0.1173	-0.0846	-0.0450	-0.4229
	(0.0096)	(0.0103)	(0.0117)	(0.0120)	(0.0139)	(0.0186)
4	-0.2266	-0.1985	-0.1809	-0.1593	-0.1018	-0.7108
	(0.0092)	(0.0103)	(0.0117)	(0.0123)	(0.0155)	(0.0292)
5	-0.3480	-0.3004	-0.2813	-0.2703	-0.2402	-1.7678
	(0.0106)	(0.0128)	(0.0128)	(0.0142)	(0.0166)	(0.0746)

Lower deciles, and those in wider spread quintiles, generally failed to produce meaningful returns under these conditions, emphasizing the critical role of liquidity in execution.

Impact of Earnings

Firms with upcoming earnings releases tended to exhibit higher short-term implied volatility, which inverts the volatility term structure. But this did not result in better returns, likely because the event volatility of earnings is well understood and accurately priced in across the term structure. Portfolios excluding earnings releases performed better, indicating that returns are driven more by structural volatility differences (the forward factor) than by earnings-related volatility spikes.

Table 7 compares returns for portfolios with and without earnings releases.

Table 7: Forward Volatility Asset Portfolio Returns

Accounting for Earnings Announcements

Each month, portfolios of forward volatility assets (FVA) are formed based on the implied volatility differential between the two legs, as defined by $\frac{1mIV}{FV(1,1)} - 1$. The FVA consists of a long two month ATM equity option straddle and a short one month ATM equity option straddle. Portfolio 1 (5) holds FVAs whose implied volatility curves have steepest upward (downward) slope. The FVAs are constructed such that the notional amounts of the one month and two month options are equal. The sample is split into two sub-samples based on whether the underlying firm releases an earnings announcement over the holding period. The period examined spans from January, 1996 to August, 2015, and includes 284,984 FVAs and 6,799 equities.

	Term Structure					
	1 (depressed)	2	3	4	5 (inverted)	5-1
Earnings	0.0047	0.0903***	0.1076***	0.1865***	0.3329***	0.3444***
	(0.0174)	(0.0242)	(0.0213)	(0.0175)	(0.0185)	(0.0213)
N	9958	13596	16555	19270	24906	
Sharpe Ratio						3.65
No Earnings	0.0761***	0.1482***	0.1941***	0.2688***	0.5081***	0.4358***
	(0.0090)	(0.0116)	(0.0119)	(0.0128)	(0.0213)	(0.0204)
N	46948	43447	40487	37773	32044	
Sharpe Ratio						4.82
Earnings -						-0.0972***
No Earnings 5-1						(0.0283)
Sharpe Ratio						-0.79

Impact of Company Size

Company size plays a significant role in the profitability of this strategy. Smaller companies tend to exhibit higher returns due to their greater sensitivity to term structure dynamics. These companies often experience sharper differences between spot and forward volatility levels, which amplify the forward factor signal. Additionally, smaller companies may have less efficient pricing, allowing for larger opportunities to exploit inverted volatility term structures.

Optimal Trade Summary

The best trades occur when a security meets all four criteria:

1. **Decile 9 or 10** (high forward factor): Represents the steepest inverted volatility term structures.
2. **Top 20% most liquid securities** (Spread Quintile 1 for narrowest bid-ask spreads): Ensures transaction costs are minimized, preserving profitability.
3. **No upcoming earnings release**: Excluding securities with earnings avoids the volatility spikes and associated risks that could distort the forward factor signal.

4. **Smaller companies**: Smaller firms often have higher returns due to their greater sensitivity to term structure dynamics and potentially higher forward factors.

This combination ensures the best opportunities by targeting trades with high returns, low transaction costs, reduced earnings-related risks, and the added potential from smaller companies' sensitivity to term structure dynamics. Keep in mind that finding a perfect fit for this is fairly rare, because bigger companies tend to be the ones that are more liquid. To ensure that we can get enough trades on, we typically look for a blend of these with higher priority on liquidity than company size. That is why a big part of this strategy is being on the hunt for good opportunities and knowing how to make sure the fills you get are actually trading a forward factor greater than the 0.16 cutoff for the 9th decile.

Now that we have covered the theory of the strategy, it's time to talk about how to actually find trades!

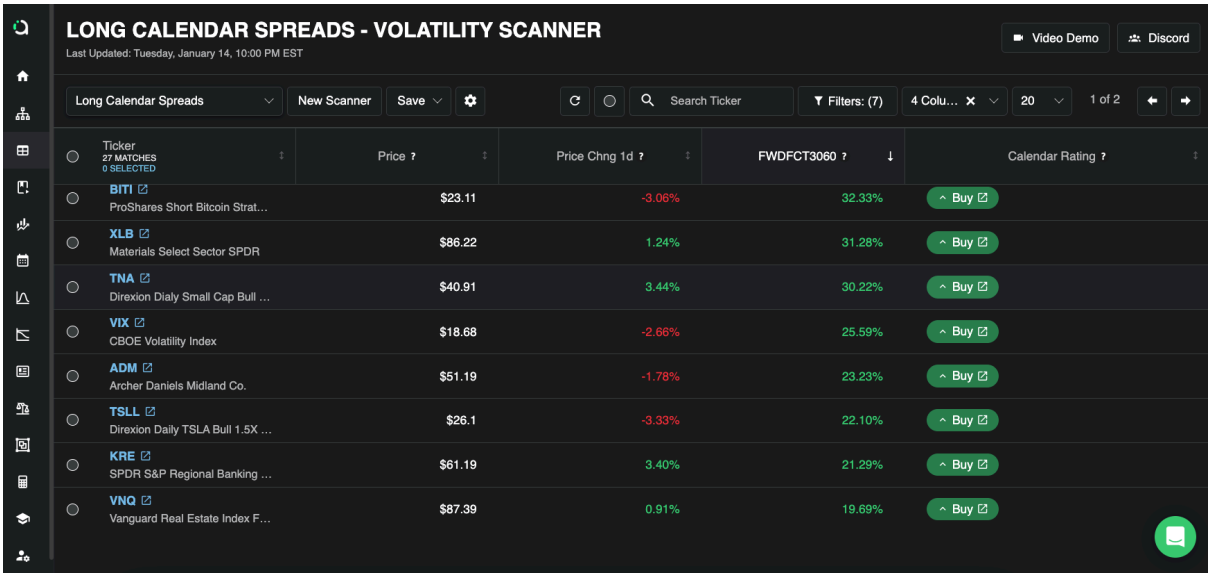
Part 4: How to Run The Strategy

Now that we have covered how calendars work, and the research behind this strategy, I am going to show you the exact steps that you take to find and place a trade for the Calendar Spread Strategy.

Step 1: Create a List of Potential Opportunities

When you are just starting off you can do this by using the “Long Calendar Spreads” scan in the Predicting Alpha Terminal. This is a basic scan designed to help you create a list of opportunities that meet the strategy criteria. It focuses on the 30DTE/60DTE calendars.

I signed into the platform, used the dropdown in the top left of the scanner to switch to the Long Calendar Spreads scan, and this is the list of trades that appeared.



There were multiple trades from this list which I placed, but the one that we are going to be analyzing together is ADM, which you can see in roughly the middle of the list.

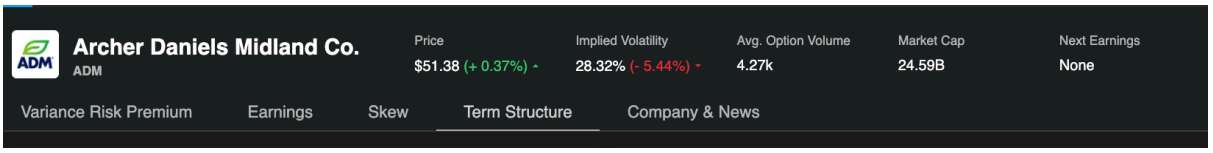
The forward factor for ADM is shown at 23.23%, which I like to see because it’s high enough above the 16% mark that I have some “wiggle room” when it comes to getting fills to still be selling a forward factor that is above the 9th decile in the research (the 16% line). Remember, it’s all about the actual fill that we can get, so having some room to manoeuvre is valuable.

Step 2: Analyze the Data

Liquidity & Earnings Check

The first thing I always check is the liquidity. Seeing an average option volume around 4,000 contracts daily gives me some confidence that there should be decent liquidity on the option chain, especially since we are trading at the money strikes. You can see the option volume at the top of the page when you look up a ticker, right beside the Implied Volatility.

I also take note that there is no scheduled earnings, (also available on the top bar, “Next Earnings”) which means it’s unlikely that there will be one in the front month expiration I sell. We wouldn’t want to take the trade if the forward factor was driven by the earnings volatility, since this is a known part of the volatility structure for a ticker and is properly priced into the option chain (covered in the previous research section).



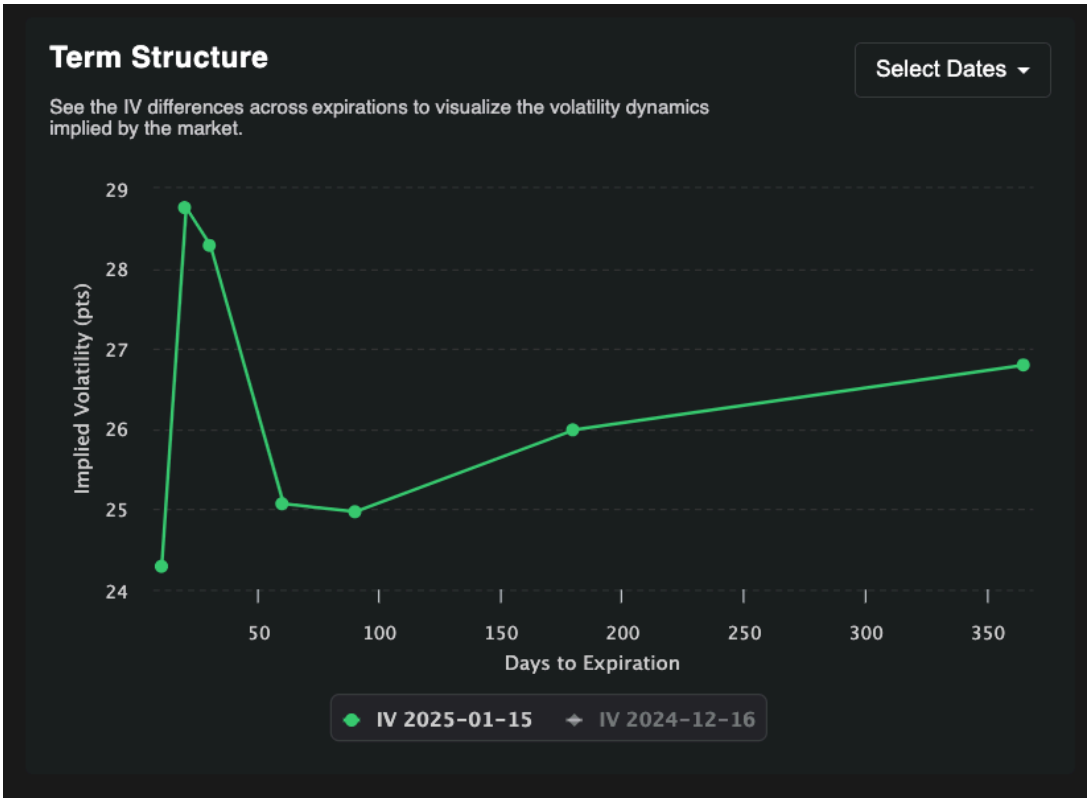
Seeing adequate liquidity and no earnings event signals to me that it’s worth continuing with my analysis.

Forward Factor Analysis

Looking at the forward factor tracker, which is found on the Term Structure page under Ticker Dashboards, we can see that the forward factor is elevated relative to normal levels. Using the bar chart on the right, we can see that the forward factor is not due to event volatility, otherwise there would be a difference between the green and blue bars (green bar uses regular implied volatility, blue uses non-event implied voaltiltiy. If there was an earnings event being priced in, you would see the blue bar being lower than the green bar).

Term Structure Analysis

When we look at the term structure, we can actually see the shape of it is in line with the effect we are seeing.

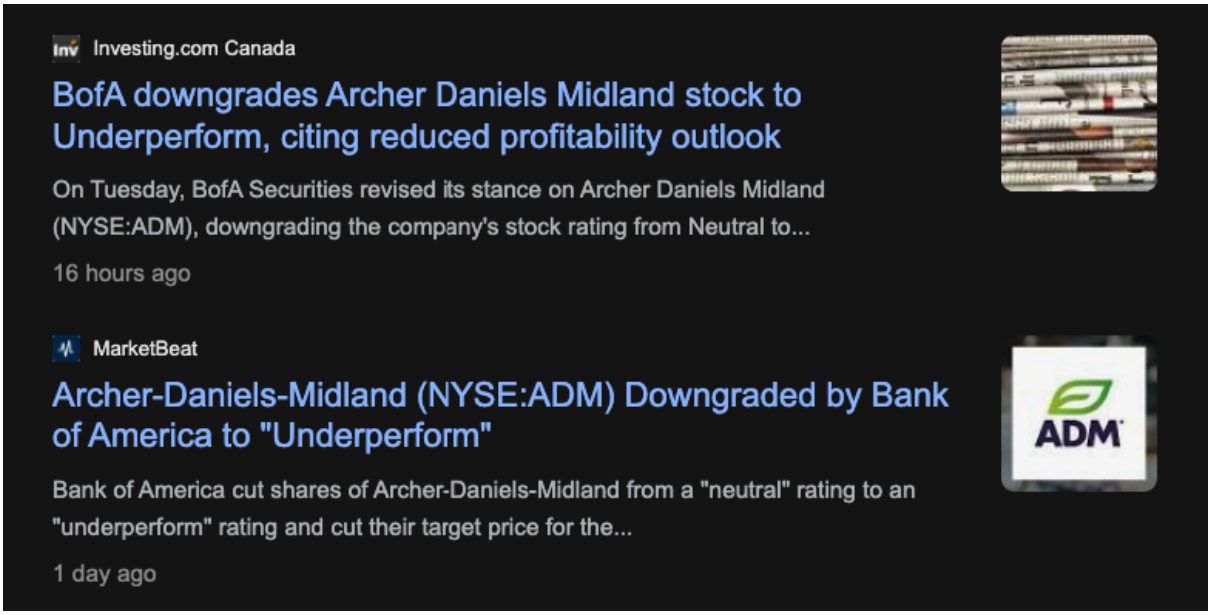


The 30 day volatility is higher than the 60 day by about 4 or 5 implied volatility points. This shape of the term structure is what is driving the high forward factor.

Step 3: Check What is Driving This Forward Factor

Sometimes there won't be an explicit event or news pending that will drive this, but in order for us to gain a clear understanding of the trade it's worth doing a quick scan of the news to see if there's anything that could cause changes to the volatility for ADM.

A quick search showed how the ticker has been underperforming recently, driven by concerns about accounting fraud tied to earnings over the past few quarters, which even led to the CEO's resignation. Bank of America recently downgraded the stock, reflecting these issues. This has understandably heightened short-term volatility relative to long-term volatility.

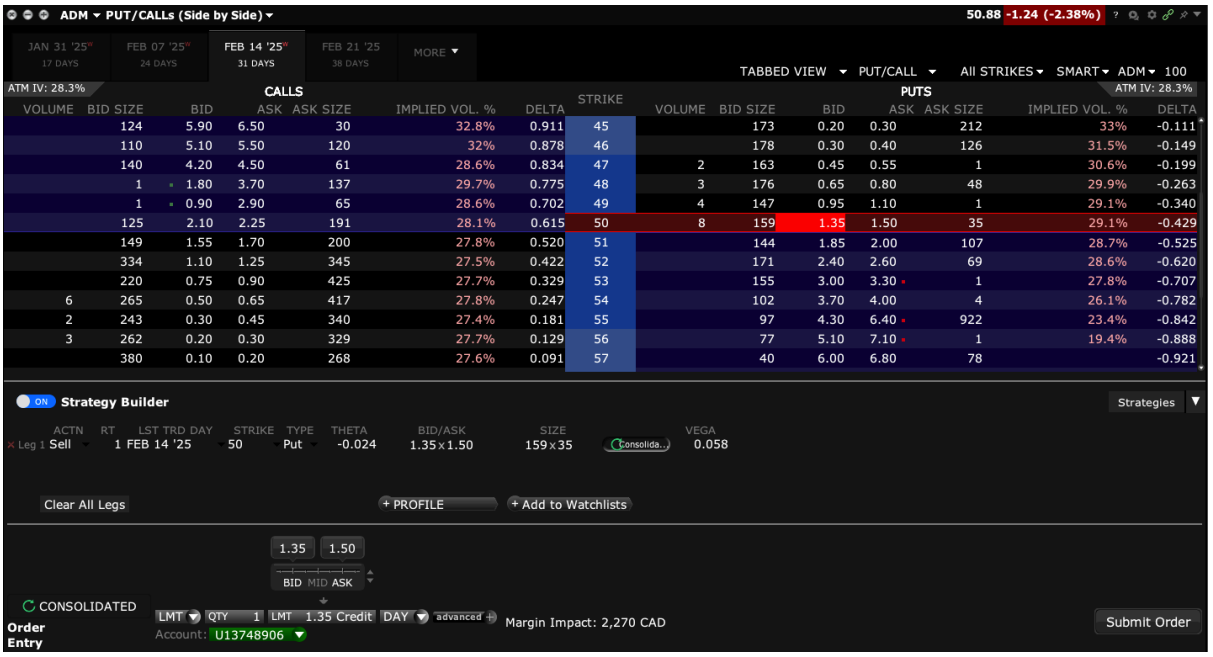


As you can see, with the stock down over 15% in the last few months + negative news, it makes sense that short term volatility is up as the stock could be expected to move lower. It makes sense why we are seeing an elevated forward factor, which is good!

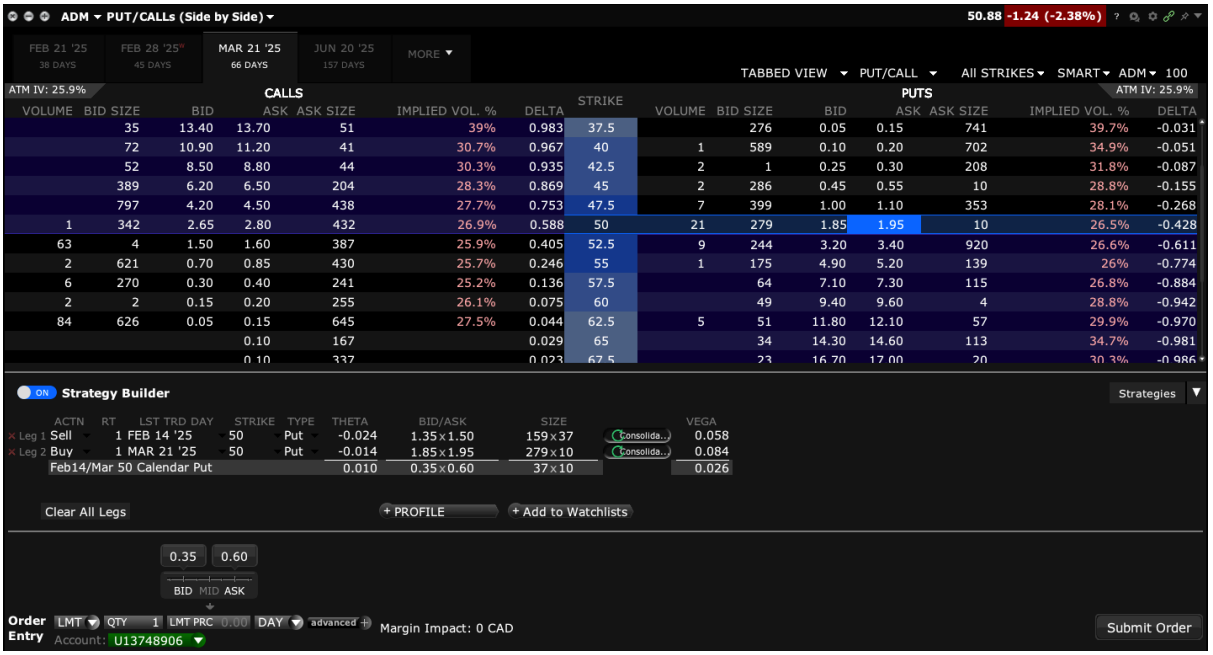
Step 4: Pick Your Live Strikes and Expirations

Now that we feel confident in placing a trade on this ticker, it's time to go and see if we can actually structure a trade like this and what price we would need to get a fill at in order to be selling this forward factor.

First lets take a look at what strike and expiration to pick for our short strike. There isn't exactly a 30DTE expiration available, so I opted for the closest one which is 31 DTE. On this expiration, the 50 strike is what I will look to sell. The reason I chose this over the 51 strike (even though the underlying is closer to 51 right now) is because I took a peek at the longer dated expirations and there is only a 50 and 52.5 strike available. I am choosing to prioritize having the same strike over the slightly more delta neutral starting point. I was indifferent about choosing call or put side in this trade so I went with put side.

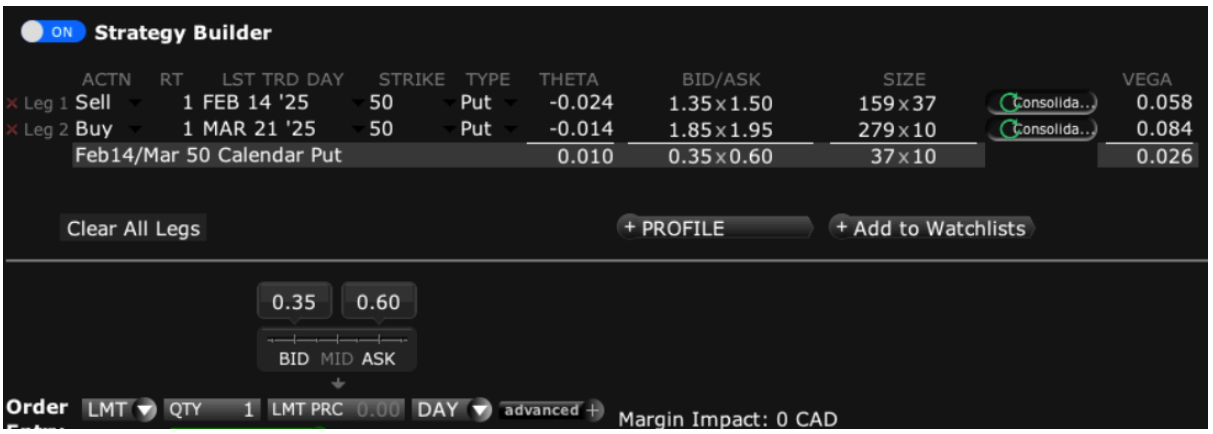


Now it's time to buy the back month long put. Looking at the live expirations, I can see there is a 66 day expiration available. This is the closest to the 60 DTE we were originally looking at, so I chose to go with it. Taking a look at the 50 put, I can see that the bid ask spread is pretty tight, so I am happy to trade this expiration. I selected the 50 put and my position is now "structured".



Step 5: Price The Forward Factor

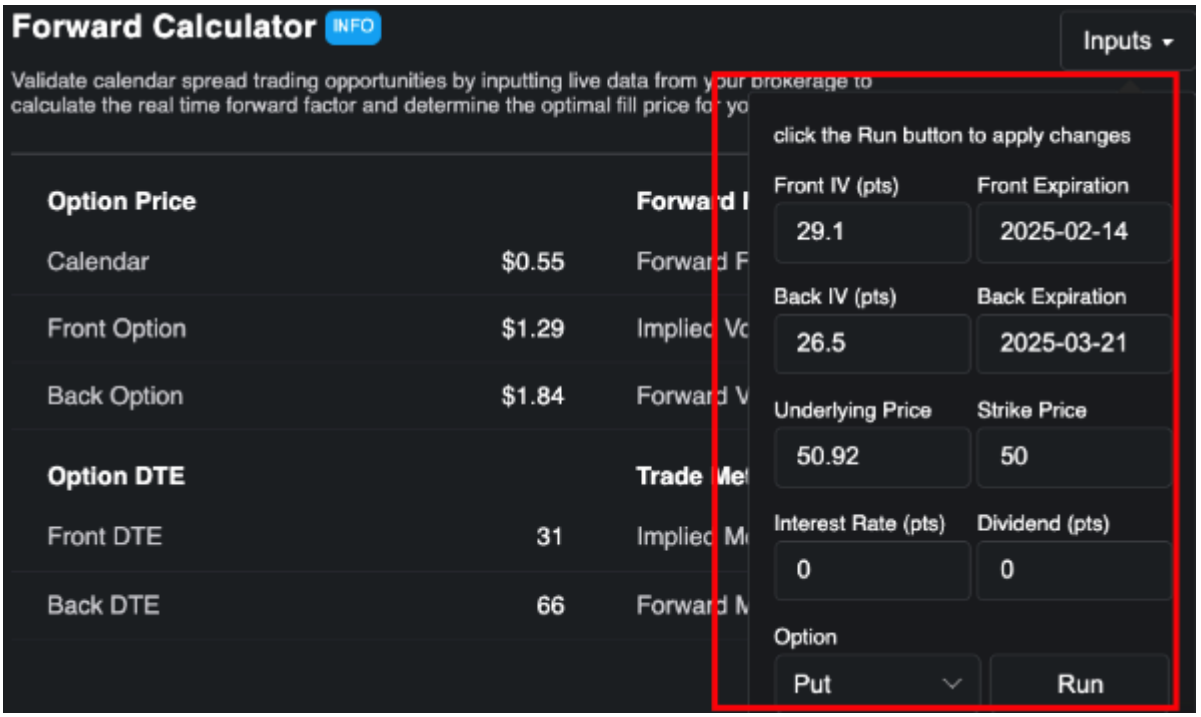
Now that we know the actual strikes, expirations, and roughly the fill we could be looking at for the trade, we can take this “live” information, put it back into the **Forward Factor Calculator** in the Predicting Alpha Terminal (found on the term structure page) and see what price we should aim to get filled at!



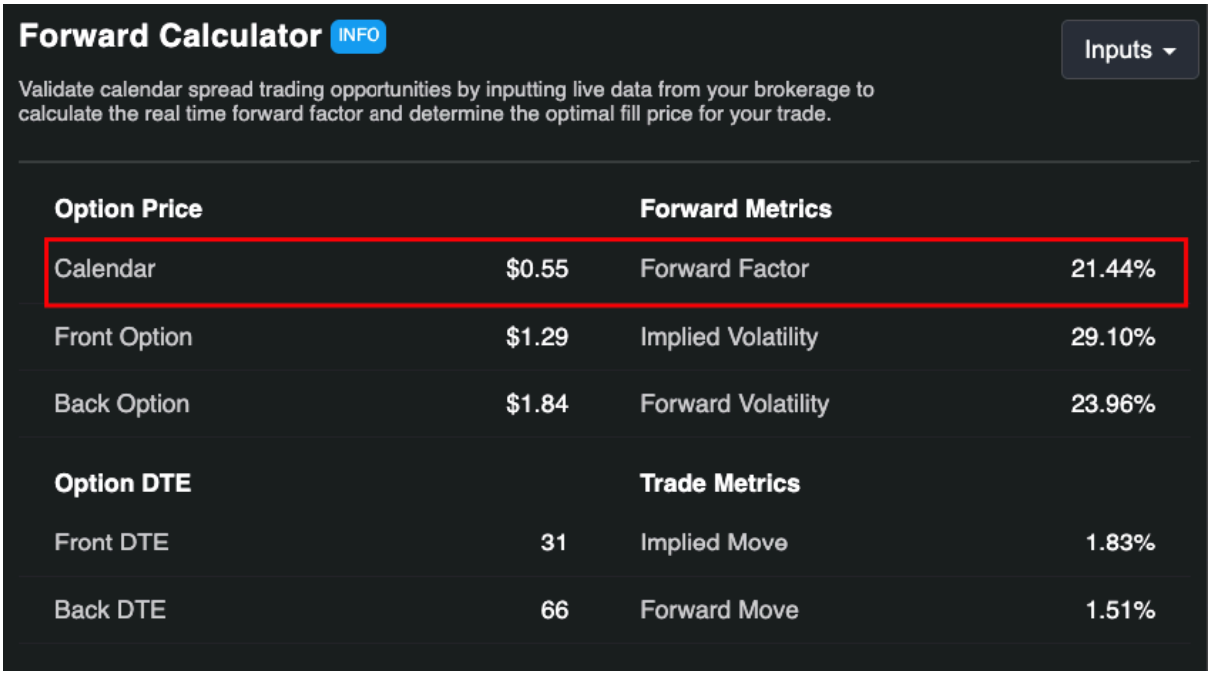
Looks like we are going to get a fill somewhere between 0.35 and 0.60. That’s the bid/ask on this trade.

Let’s plug some of these numbers into the calculator to see what price we could be selling at to try and get a fill above the 16% forward factor (and hopefully even higher than that!).

Here are the calculator inputs I used. This is all data from the brokerage. I used the implied volatility numbers for each of the options that was quoted to me on the option chain.



When I click run, here's what the output looked like:



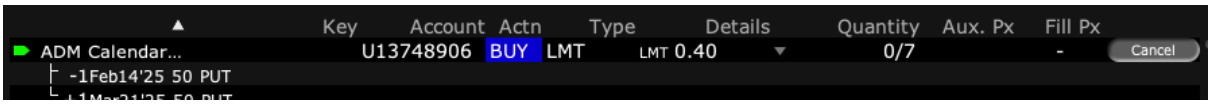
The main numbers that I am concerned with are the calendar price and the forward factor. This can be interpreted as follows:

"If I can buy the calendar spread for $0.55 , I am locking in the trade at a forward factor of 21.44%".

Realistically, this is what I should be aiming for, and since it's at the higher end of the bid/ask spread I am pretty confident I can get this fill. So let's give it a try now!

Step 6: Get a Fill On The Trade!

I decided to start by trying to get a better fill than $0.55. I started near the bid, at $0.40, and left the order sitting for a while.



No luck! Looks like I will have to move up my bid.

I eventually did get a fill, exactly at $0.55.

Step 7: Managing The Trade

I now have this trade on. It is one of many calendar spreads that should be in the portfolio at any given time. Remember, we are playing a game of small edges and it requires a large sample size for the expected value to really show itself consistently.

I will be monitoring the forward factor of this trade and where the underlying stock moves.

Delta Hedging Requirements

As the trade progresses, you will need to manage your delta. There are two ways that we do this.

1. If I can still buy calendars at an elevated Forward Factor, I will open up new calendar spreads using the same expirations, but with different strikes to balance out my delta exposure. This increases my overall exposure to the trade, but since it is a debit position I don't run the same margin risks as I do with other strategies we run. So I am comfortable managing my delta this way and increasing my overall exposure to the trade.

2. I can hedge with shares, similar to traditional delta hedging. This does not increase my overall exposure to the trade and still accomplishes the goal of returning us to delta neutrality.

Closing For a Profit

I will typically hold this trade for a profit until one of two things occur:

1. The front month option is about to expire
2. The reason the forward factor was elevated is resolved

Closing For a Loss

I will typically hold this trade for a loss until one of two things occur:

1. There has been a massive move and we no longer have exposure to the difference in implied volatility between the expirations we sold.
2. The reason the forward factor was elevated is resolved.

To sum it up, we are typically holding until the front month expiration unless the event causing the term structure to be in severe backwardation is resolved or there is a massive move in the underlying.

Diversification and Placing Many Trades

Since each of these trades are placed on different underlyings with different reasons why their Forward Factors are elevated, the outcomes for each trade are mostly uncorrelated, which means there is benefit in diversification.

Always remember that the point is not to find one calendar spread. The goal is to put together a portfolio of trades that all share the commonality of having a Forward Factor greater than 16%. This is no different from how we run any of our other strategies in the Predicting Alpha Community. We accept that we have a small edge, and to help it play out we intentionally look to place many trades in small sizes.

Conclusion

Now that we've covered everything you need to know about running the Calendar Spread Strategy, you're ready to start implementing it in your portfolio. The key to success with this strategy lies in careful execution, proper trade sizing, and managing trades effectively. So, dive in and let me know how things go! I'm here to help if you need support along the way.

Happy trading,

Sean